

BGSC Platform — Complete Feature Specification & Architecture

Table of Contents

1. Application Overview
 2. Architecture & Technical Stack
 3. Global UI/UX Frame & Navigation
 4. Data Models (Domain-Driven)
 5. Page & Feature Specifications
 6. Popups & Modal Specifications
 7. Administration & RBAC
 8. Event-Driven System Design
 9. Third-Party Integrations
 10. Notification System
 11. Security, Privacy & Moderation
 12. Settings & Preferences
 13. Search & Discovery
 14. Analytics & Success Metrics
 15. Media & Content Specifications
 16. Future Roadmap
 17. Phased Development Roadmap
 18. Appendix: Glossary
-

1. Application Overview

1.1 Introduction

The platform is a comprehensive, **event-driven** application utilizing a microservices architecture, designed to serve as the digital hub for the BITS Goa sports and esports communities. It acts as a dual-purpose platform: a public-facing social and event engagement hub for students and gamers, and a robust internal workspace (“Union Page”) for the core organizing committees (BGSC, BGEC, FitSoc).

By integrating external activity trackers (Strava, Steam), gamifying community participation through a points system, and providing seamless event management, the platform unifies campus recreation. Furthermore, it empowers the internal crew with advanced task management, automated communication, and role-based access controls to efficiently execute large-scale leagues and events.

1.2 Target Users

- **Guest:** Unauthenticated visitors with read-only access to public events and announcements.
- **User:** Authenticated students with access to personalized feeds, social features, and event registration.
- **Member (Union):** BGSC/BGEC crew members with restricted internal access to assigned events/tasks.
- **Core:** Event and workspace managers with full operational access to events they are assigned to.
- **Coordinator:** Top-level operational managers with full access except Founder role modification.
- **Founder / Admin:** Absolute access to all system configurations, roles, and data.

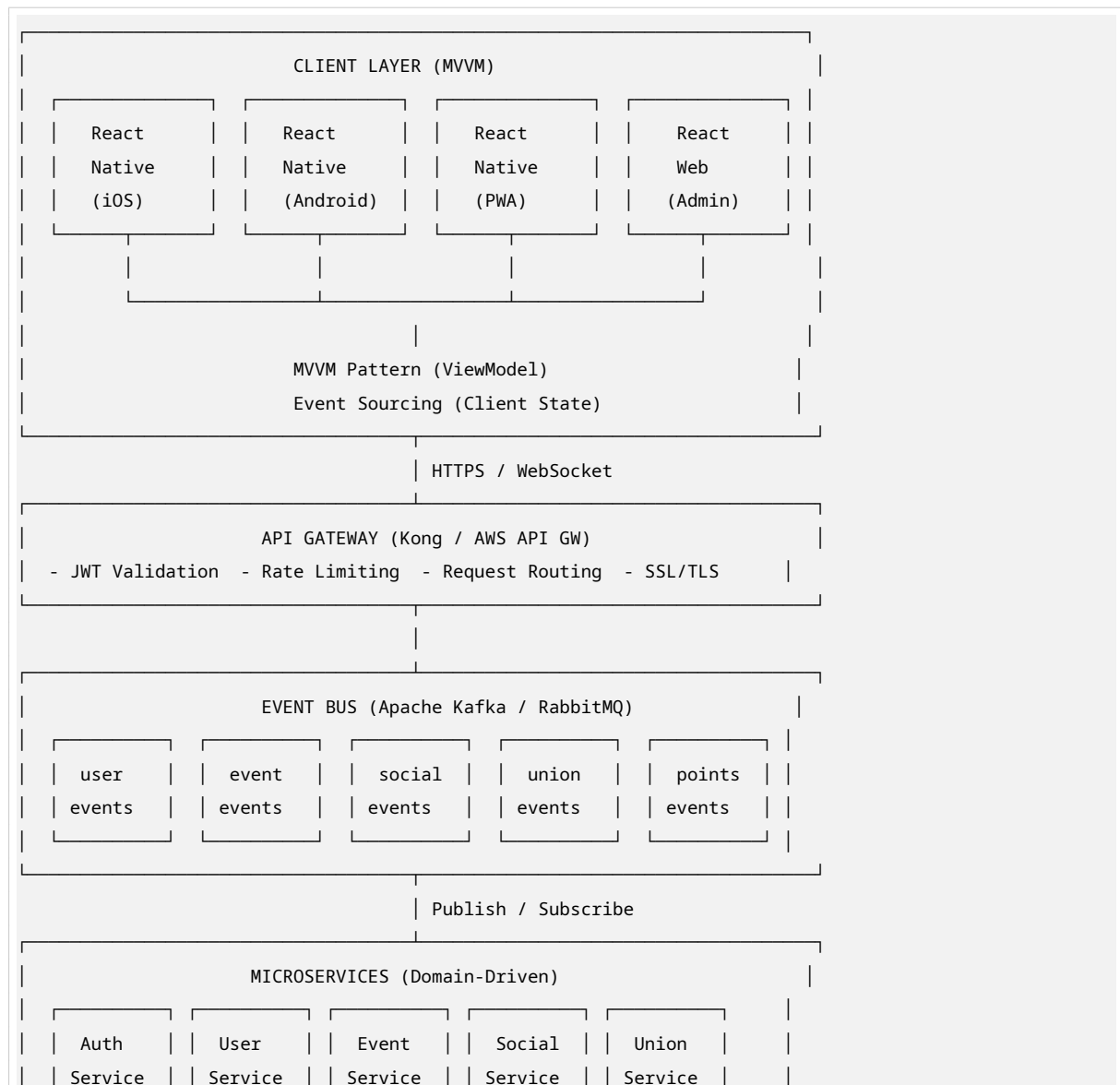
1.3 Core Philosophy

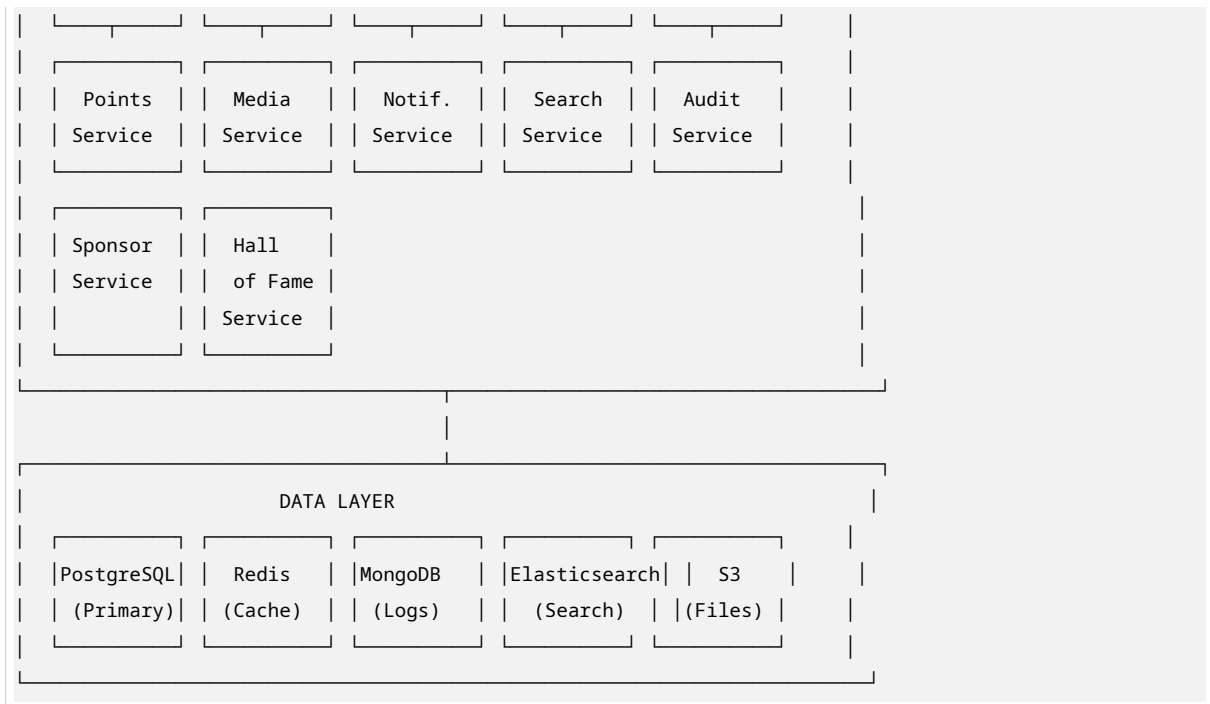
- **Events First:** The platform revolves around event discovery, registration, and management.
- **Gamified Engagement:** Points, challenges, leaderboards, and sponsor affiliations drive participation.
- **Community-Driven:** Social features, matchmaking, and content sharing create network effects.
- **Operational Excellence:** Internal tools must be as polished as public-facing features.
- **Event-Driven Architecture:** All state changes propagate through events, enabling real-time updates, audit trails, and loose coupling between services.

2. Architecture & Technical Stack

2.1 Architectural Pattern: Event-Driven Microservices

The platform follows an **Event-Driven Architecture (EDA)** where all significant state changes emit domain events. Services communicate asynchronously via an event bus, enabling loose coupling, horizontal scalability, and real-time feature support.





Event-Driven Flow Examples Event Registration Flow:

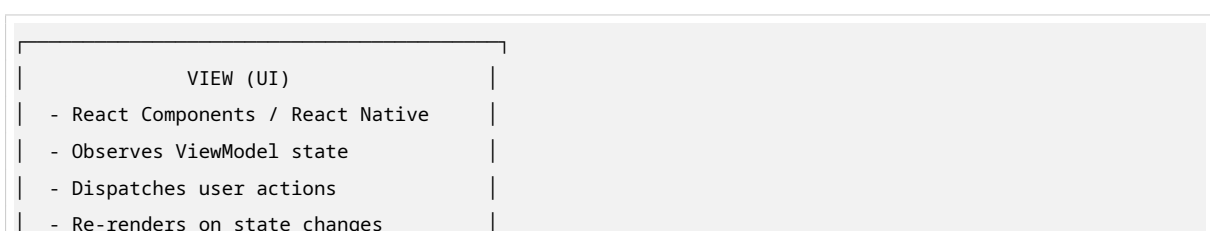
1. User clicks “Register” → POST /api/events/{id}/register
2. Event Service validates → writes to DB → emits EventRegistrationCreated event
3. Points Service consumes → awards points → emits PointsAwarded event
4. Sponsor Service consumes → updates sponsor fan counts if user is affiliated
5. Notification Service consumes → sends push/email
6. User Service consumes → updates user’s event history cache

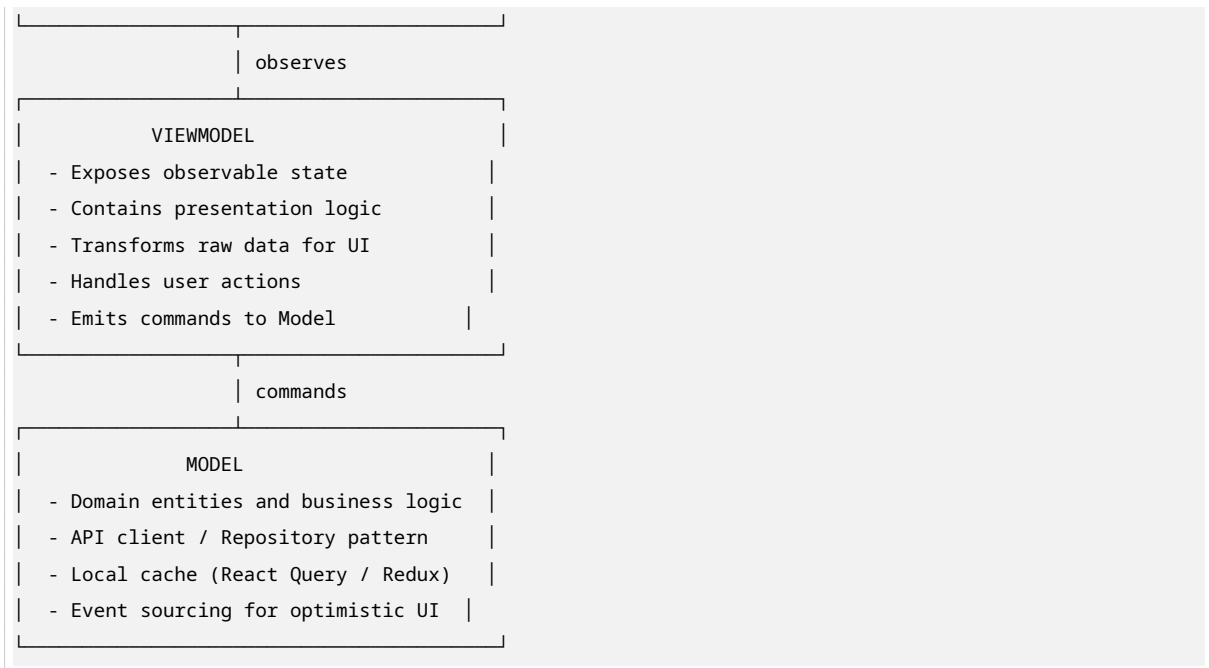
Auction Bid Flow:

1. Captain places bid → POST /api/auctions/{id}/bid
2. Auction Service validates (purse, timing) → writes to DB → emits AuctionBidPlaced event
3. WebSocket Gateway consumes → broadcasts to all connected clients in auction room
4. Notification Service consumes → updates other captains
5. Audit Service consumes → logs bid for transparency

2.2 Frontend Pattern: MVVM (Model-View-ViewModel)

The client application follows **MVVM (Model-View-ViewModel)** pattern with **Event Sourcing** for local state management.





State Management Strategy

- **Global State:** Zustand / Redux Toolkit (auth, user, notifications)
- **Server State:** React Query / TanStack Query (events, posts, friends)
- **Local State:** React useState / useReducer (forms, UI toggles)
- **Event Store:** Client-side event log for optimistic updates and offline support

MVVM Implementation Example (Event Registration)

```
// Model (Repository)
class EventRepository {
  async registerForEvent(eventId: string): Promise<Registration> {
    const response = await api.post(`/events/${eventId}/register`);
    return response.data;
  }
}

// ViewModel
class EventViewModel {
  registrationState = observable<AsyncState<Registration>>({ status: 'idle' });

  async register(eventId: string) {
    this.registrationState.set({ status: 'loading' });
    try {
      const registration = await eventRepository.registerForEvent(eventId);
      this.registrationState.set({ status: 'success', data: registration });
      eventBus.emit('REGISTRATION_SUCCESS', { eventId, registration });
    } catch (error) {
      this.registrationState.set({ status: 'error', error });
    }
  }
}
```

```
// View (React Component)
const EventCard = ({ eventId }: { eventId: string }) => {
  const vm = useEventViewModel(eventId);
  const state = useObservable(vm.registrationState);

  return (
    <Card>
      <Button
        onPress={() => vm.register(eventId)}
        loading={state.status === 'loading'}
      >
        {state.status === 'success' ? 'Registered!' : 'Register'}
      </Button>
    </Card>
  );
};
```

2.3 System Stack

Layer	Technology	Rationale
Mobile App	React Native (Expo)	Single codebase for iOS/Android; campus users are mobile-first
Web Frontend	React + Tailwind CSS	Admin dashboards, PWA for desktop users
Frontend Pattern	MVVM + Event Sourcing	Clean separation, testable, real-time reactive UI
Backend API	Node.js (NestJS)	Native TypeScript, excellent event-driven support, high concurrency
Event Bus	Apache Kafka	Durable event log, replay capability, horizontal scaling
Primary Database	PostgreSQL	Relational data integrity, ACID compliance for transactions
Cache & Sessions	Redis	Session store, rate limiting, real-time leaderboards, pub/sub
Document Store	MongoDB	Media metadata, event logs, unstructured content
Real-Time	Socket.io + Redis Adapter	Auctions, chat, live leaderboard updates across server instances
Message Queue	BullMQ (Redis-based)	Async job processing: WhatsApp sends, email digests, image processing
File Storage	AWS S3 / Cloudflare R2	Media, PDFs, profile pictures, exports
CDN	CloudFront / Cloudflare	Global delivery for static assets and media
Search	Elasticsearch	Fuzzy username/event search, typo tolerance, faceted search

Table 1 – continued

Layer	Technology	Rationale
Authentication	JWT (Access + Refresh) + OAuth2	Stateless, scalable auth; Google OAuth integration
Push Notifications	Firebase Cloud Messaging (FCM)	Cross-platform push delivery
Error Tracking	Sentry	Real-time error monitoring and alerting
Infrastructure Monitoring	Grafana + Prometheus	API latency, DB performance, queue depth
Container Orchestration	Docker + Kubernetes	Scalable microservices deployment
CI/CD	GitHub Actions + ArgoCD	Automated testing and deployment

2.4 Scalability Considerations

- **Horizontal Pod Autoscaling:** API servers scale based on CPU/memory and request queue depth
- **Database Read Replicas:** Event browsing and leaderboard queries served from replicas
- **CDN Caching:** Public event pages and media cached at edge locations
- **Async Job Processing:** Image resizing, video compression, bulk notifications via worker queues
- **Event Replay:** Kafka event log enables replay for debugging, new service hydration, and audit
- **CQRS (Command Query Responsibility Segregation):** Write model (PostgreSQL) separate from read model (Redis + Elasticsearch) for complex queries

3. Global UI/UX Frame & Navigation

3.1 Dynamic Status Bar

A persistent, context-aware status bar present on all screens:

- **Center:** Contextual Logo that changes based on the current module:
 - Home / Landing → BGSC logo
 - Esports Events → BGEC logo
 - Fitness Events → FitSoc logo
 - Specific Events (e.g., Offside) → Event-specific logo
- **Left:** Global Navigation / Side Drawer toggle containing routing links to all major modules.
- **Right:**
 - Authenticated: User Profile Picture (triggers Account Actions pop-up).
 - Guest: Login button (routes to Login/Registration Page).

3.2 Navigation Drawer (Side Drawer)

Contains links to:

- Home / Landing Page

- Point System and Challenge Page
- Sponsor / Newsletters Page
- Friends Page
- Events Page
- Leaderboards
- Hall of Fame
- Store Page
- Media Page
- Feedback and Contact Us
- Union Page (internal-only, visible based on role)
- Users Page (admin-only, visible based on role)

3.3 Authentication States

- **Guest View:** Mirrors the core features of the public BGSC website. Read-only access to public events, announcements, and public posts. No social interactions.
- **Authenticated View:** Unlocks personalized feeds, friend systems, internal routing, user profiles, points system, and event registration.

4. Data Models (Domain-Driven)

4.1 Core Entities

```
User: id, username, email, password_hash, contact, role, avatar_url,
      interests[], socials{}, strava_id, steam_id, points_balance,
      status, created_at, last_active, settings{}, newsletter_subscriptions[],
      active_sponsor_id

Event: id, title, description, type[LE|DE|ALL|DLL], status[upcoming|ongoing|past],
       start_date, end_date, venue, rules_pdf_url, award_list[],
       needs_leaderboard, is_teamed, team_size, max_teams,
       registration_deadline, created_by, tags[],
       core_admins[], points_pool{}, is_auction_based

Team: id, event_id, name, captain_id, members[], status[open|invite_only|closed],
      registration_cost, max_members, invite_code

Task: id, title, description, type[quick|standard|pathway|event_task],
      assignees[], deadline, priority, status[active|abandoned|completed],
      parent_task_id, event_id, created_by, updates[], reminders[],
      is_public, chat_group_id, steps[]

Post: id, user_id, media_urls[], caption, tags[],
      visibility[public|protected|private], likes_enabled, comments_enabled,
      comments_visibility[public|protected|private], shares_allowed, created_at
```

Friendship: id, requester_id, recipient_id, status[pending|accepted|rejected|blocked], created_at

Notification: id, user_id, type, title, body, data_payload,
 is_read, channel[in_app|push|email|whatsapp], created_at

PointTransaction: id, user_id, amount, type[earn|spend|refund],
 source[event|challenge|store|leaderboard], reference_id, created_at

Announcement: id, title, body, type[BGEC|FitSoc|Airball|Offside|PowerPlay|AroundTheNet|Deuce|Highlight|↔ Teams],
 tags[], created_by, whatsapp_sent, created_at, expires_at

Match: id, event_id, team_a_id, team_b_id, score_a, score_b,
 parameters{}, status[scheduled|ongoing|completed|cancelled],
 scheduled_at, completed_at, venue

Auction: id, event_id, player_id, current_bid, current_bidder_id,
 status[upcoming|active|sold|unsold], min_bid_increment,
 bid_history[], timer_start, timer_end

StoreItem: id, name, description, image_url, points_cost, stock,
 category[merch|indie_game], status[available|sold_out|discontinued]

Challenge: id, title, description, domain, team_limit, time_limit,
 resource_links[], award_points, difficulty[easy|medium|hard|legend],
 status[active|completed|archived]

MediaAlbum: id, event_id, title, cover_image_url, media_items[],
 visibility, created_at

FeedbackTicket: id, user_id, category[bug|feature|complaint|general],
 severity, description, attachments[], status, created_at, response

AuditLog: id, actor_id, action, target_type, target_id,
 previous_value, new_value, timestamp

Sponsor: id, name, logo_url, description, website_url,
 tenure_start, tenure_end, status[active|inactive],
 prizes[], total_fans, ranking

UserSponsorAffiliation: id, user_id, sponsor_id,
 affiliated_at, fan_count, events_won[],
 total_points_contributed

SponsorRanking: id, sponsor_id, semester, year,
 total_fans, rank, events_won_count,
 top_contributors[], prize_awarded

SponsorPrize: id, sponsor_id, title, description,
 criteria[top_fans|most_wins|highest_rank],


```
threshold, points_cost, status[available|claimed|expired]

SponsorMedia: id, sponsor_id, media_url, media_type[video|image],
               category[hero_video|ad|story|promo|archive],
               title, description, duration,
               auto_play, loop, mute_default,
               cta_text, cta_link,
               display_order, status[active|archived],
               created_at, expires_at
```

5. Page & Feature Specifications

5.1 F: Login / Registration Page

Visibility: Public (Guest + All authenticated users for re-login)

Common Fields (Both Login & Register):

- Forgot password link
- Sign up with Google (OAuth2)

Login Page:

- Move to register page link
- Username / Email *U (required, unique)
- Password * (required)
- Session persistence toggle (“Keep me logged in”)

Registration Page:

- Username *U (required, unique)
- Email *U (required, unique)
- Contact (optional)
- Password * (required)
- Re-enter Password * (required, must match)
- Terms of Service and Privacy Policy acceptance checkbox *
- **Sponsor Selection:** Dropdown of active sponsors for the current semester/year. User must select one *. This becomes their affiliated sponsor.
- Post-registration: Auto-triggers Get Started PopUp sequence

5.2 F: Home Page (Landing Page)

Visibility: Public (Guest + Authenticated) **Layout:** Tab navigation with three sub-pages.

Tab 1: Introduction / Landing

- BGSC and subsection (BGEC, FitSoc) introduction
- Catchy one-liners and campus sports culture highlights
- **“What Our Heads Have to Say” Section:**
 - Pixelated/photographic portraits of coordinators posed in announcement stance
 - Thought-box / comic-style speech bubbles displaying their latest announcement
 - If a coordinator has no announcements, a meme is displayed instead
 - Quick link to full Announcements page
 - Reference design: <https://www.behance.net/gallery/27476403/Minions-Website>

Tab 2: Announcements

- Centralized announcement feed (similar to Moodle/AWS broadcasting)
- WhatsApp API integration for automatic broadcasting (budget-dependent)
- Coordinator and Admin access to create new announcements via Make Announcement PopUp
- Retention policy: Only announcements from the past 4 months are displayed and stored
- Attribution: Shows which coordinator/admin made each announcement
- Categories: BGEC, FitSoc, Airball, Offside, PowerPlay, Around The Net, Deuce, Highlight Events, Teams

Tab 3: General Social Feed

- Public posts from friends and community (public visibility only)
- Add Post button (floating action button, bottom-right)
 - Guest click → redirect to Login Page
 - Authenticated click → opens Add Post PopUp

5.3 P: User-Profile Page

Visibility: Authenticated only (replaced by Login button if logged out) **Status Bar:** Custom status bar for this page:

- Left: Back button → returns to Landing Page
- Center: “Account Actions” button → opens Account Action PopUp
- Right: Profile Picture → opens Profile Picture section

Player Card / User Card (Center of Page) Inspired by: <https://pin.it/81Wcd43Gj>

- Username
- Avatar (2D illustration)
- Short Description / Bio
- Sports / Esports Interests (Games)
- Animations (entrance/idle)
- Goal Achievements
- Custom Tags (game/sport-specific, e.g., “Striker”, “IGL”)
- **Active Sponsor Badge:** Logo and name of currently affiliated sponsor

- Social Media Links / Handles
- Fixed Rating Section (computed metrics)
- Shareable Card export (image generation for social media)
- Matchmaking integration: Used elsewhere to find similar players

User Info Section (Below Card)

- Tags given by friends
- Name, Email, Contact
- Interest fields (Sports + Esports)
- Newsletter subscriptions and interaction data
- Sport activity (Strava integration)
- Esport activity (Steam integration)
- Connected social accounts
- **Sponsor Stats:**
 - Current sponsor name and logo
 - Personal fan count earned for sponsor
 - Events won contributing to sponsor
 - Sponsor ranking position

Event Suggestions Section

- Upcoming and highlight events tailored to user interests
- Forces discovery of new categories (exploration algorithm)
- For teamed tournaments: list of open public teams below each event card

Friends Suggestions Section

- Suggestions based on contact list, shared interests, or event activity

History Section

- Events participated in
- Games played / match history
- Challenge completions
- Sponsor contribution history (events where user earned fans for their sponsor)

Future Roadmap

- “Looking For?” dating/matchmaking feature (relationship status)
- Deeper connection features based on community interests

5.4 P: Friends Page

Visibility: Authenticated only **Layout:** Tab navigation with 5 tabs.

Tab 1: General Chats & Search

- Friend list with chat previews
- Search bar:

- Find friend by username/name
 - Search results show: Mutuals, Activities/Interests, Player Card preview, Sponsor affiliation
- Friend Requests button (beside search bar):
 - View all incoming/outgoing requests
 - Accept / Reject / Cancel
 - Add / Remove friend actions
- Chat Section:
 - Direct Messages (DMs)
 - Group Chats
 - Community Servers
- Active Users indicator (online status)

Tab 2: Activities & Events

- Events that friends have participated in
- Filter by live / past

Tab 3: Recent Achievements

- List of friends with recent wins/achievements
- Expandable detail view per friend
- Sponsor fan contributions visible

Tab 4: Challenge Friends

- Browse available challenges
- Select challenge + target friend(s)
- Send challenge invitation
- Challenge types: Physical (space/time required) or Digital (timeline-based)

Tab 5: Team Up For Event

- Select event from upcoming list
- Send team-up requests to specific friends
- View open public teams for that event

Friends System Logic

- Suggest friends based on similar interests + sponsor affiliation (optional filter)
- Weekly/Monthly interest update prompts (popup)
- Interest auto-update based on newsletter interaction
- Collaboration option for games/events
- Active friends / teammates indicator
- Friend participation tracking (live and past)
- Challenge system for both sports and esports

Future: Share Clips (Instants / Stories)

- 3-tier story system:
 - Non-Judgemental (Close circle)
 - Close (Friends)

- General (Public / All friends)
- Users can create custom groups beyond the default 3
- 24h ephemeral content

5.5 F: Events Page

Visibility: Public (browse), Authenticated (registration) **Layout:** Tab navigation.

Event Categories (Tabs)

- Leagues (Sports + Esports)
- BGEC Events
- FitSoc Events
- General Events (Highlight events, Waves, sponsored events)

Filters (Per Tab)

- Past
- Upcoming
- Ongoing
- Multi-select enabled (can view all simultaneously)

Event Details View (For All Events)

- Event info (title, description, rules PDF/link, awards)
- **Sponsor Leaderboard Preview:** If event is active, show which sponsor is leading in fan contributions for this event type
- Event registration section:
 - Name, Game name
 - Role selection: Team Captain or Team Member
 - * If Captain: Team name input
 - * Team participation count
 - * Team status: Open / Invite Only / Closed
- Event Team Formation Section:
 - View team invites / own team
 - Send team invites to users who are “Open to join”
 - Search teams with filters
 - User toggle: Open / Closed / Invite Only (controls if others can add them)
- Event Leaderboard (if active and enabled)
- Event Status indicator
- Event Results (post-completion)
- **Post-Event Sponsor Update:** When results are published, users who won earn virtual fans for their sponsor. Display: “+X fans earned for [Sponsor Name]”

League-Specific Registration

- Same as events, plus:
- If auction-based:
 - Team Captain: No additional fields
 - Team Member: Starting cost / base price input

- System displays: Average price, deviation, variance to guide pricing

Auction Event Interface (Dedicated Page)

- **Management View:**
 - List of all team captains
 - Assign auction purse to each captain ($\text{Pool} = K \times \Sigma \text{Base Prices}$)
 - Set minimum bid bracket
 - Select player to put on auction stand
 - Start/Stop auction controls
 - OC Override Quota: Can manually adjust up to 3/7ths of player base prices
- **Team Captain View:**
 - Live player on stand
 - Current bid display
 - Raise bid button (multiples of min bracket)
 - Purse remaining display
- **General User View:**
 - Live bid updates
 - 5-second countdown timer per bid
 - Bid history
- **Bid Finality:** Once placed, bid is final. After 5-second countdown, management can close the bid or let it continue.
- **Auction Closure:** Player marked as Sold or Unsold.

Event Administration (Coordinator / Admin / Core)

- Bottom-right FAB (Floating Action Button) visible only to Admin/Coordinator
- Opens New Event PopUp with custom status bar (back button only)
- Event creation categories:
 - **Needs Leaderboard:**
 - * Leagues:
 - ALL (Auction Leaderboard League)
 - DLL (Direct Leaderboard League)
 - * LE (Leaderboard Event)
 - **No Leaderboard:**
 - * DE (Direct Event)

Event Editing (Core with Assignment)

- If user is Core and assigned to event: “Add Members” option visible
- Added Members get edit access to:
 - Edit/delete/update team details/stats
 - Add new matches via visual league graph
 - Add match stats
 - Update match results
- Event auto-closes after deadline or last match completion

5.6 F: Leaderboards Page

Visibility: Public (view), Authenticated (participate/invest)

- View all active events with leaderboards
- Filter by tags, participation status, event type
- **Points Investment:** If user is a participant and event permits, user can invest points to improve rank
- Leaderboard formats supported: Round Robin, Upper-Lower Bracket, Direct Elimination, Elimination after N fails
- Min participant threshold required to activate leaderboard
- Score normalization controls (lower limit 0, upper limit 1000)

5.7 P: Point System & Challenge Page

Visibility: Authenticated only

Points System

- User points balance display
- Points earning sources:
 - Event participation
 - Challenge completion
 - Platform engagement (posts, invites)
 - Sponsor-related bonuses (winning events for your sponsor)
- Points spending:
 - Store redemption
 - Leaderboard investment
- Transaction history

Challenge System

- Challenge browser:
 - Domain filter (Sports, Esports, Game Dev, General)
 - Difficulty levels: Easy, Medium, Hard, Legend
 - Team limit indicator
 - Time limit display
 - Resource section (help links/guides)
 - Award points display
- Legend-level challenges grant Hall of Fame entry
- Challenge types:
 - Physical (dedicated space/time required)
 - Digital (timeline-based, details revealed upon acceptance)
- Progress tracking and submission portal

5.8 F: Sponsor / Newsletters Page

Visibility: Public **Layout:** Immersive media-first experience with video backgrounds and interactive ad placements.

Hero Section — Sponsor Showcase

- **Background Video Loop:** Full-screen or section-level auto-playing video showcasing active sponsors
 - Muted by default with unmute toggle
 - Looping 10–30 second highlight reels per sponsor
 - Parallax scroll effect as user scrolls down
 - Fallback poster image for low-bandwidth or reduced-motion preferences
- **Sponsor Carousel:** Horizontal swipeable/auto-rotating carousel of active sponsors
 - Each card: Logo, name, tagline, “View Profile” CTA
 - Tap to expand into full sponsor detail view

Active Sponsors Section

- **Current Tenure Display:** Active sponsors for the current semester/year with countdown to tenure end
- **Sponsor Cards:**
 - Sponsor name, logo, description, website link
 - Total fan count (aggregate from all affiliated users)
 - Current ranking among active sponsors
 - Number of affiliated users
 - Events won count
- **Sponsor Leaderboard:**
 - Ranked list of all active sponsors
 - Sort by: Total Fans, Events Won, Affiliated Users
 - Time-filter: This Semester, This Year, All Time
 - Visual bar chart showing fan distribution
- **User Affiliation:**
 - Current user sponsor badge
 - “Change Sponsor” button (limited to once per semester)
 - Fan contribution breakdown: “You have earned X fans for [Sponsor]”

Sponsor Video Ads & Promotions

- **Sponsor Ad Slots:** Dedicated video ad placements between sponsor cards
 - 15–30 second non-skippable sponsor video ads
 - Auto-play on scroll-into-view (muted)
 - Tap to unmute / expand to full-screen
 - “Learn More” CTA button overlay at end of video
 - Ad frequency cap: max 3 video ads per session
- **Sponsor Stories:** Vertical short-form video stories (Instagram Reels style)
 - 15–60 second sponsor-branded content
 - Swipe-up to visit sponsor website or claim offer
 - Full-screen immersive viewer with progress bar
- **Sponsor Livestreams (Future):** Embedded livestream player for sponsor-hosted events or product launches

Sponsor Detail View (Expanded Card)

- **Hero Video:** Full-width sponsor brand video (auto-play, muted, looping)
- **Brand Info:** Name, logo, description, website, social links
- **Media Gallery:** Horizontal scroll of sponsor images and video clips
 - Auto-play video thumbnails on hover/focus
 - Tap to open full-screen lightbox viewer
- **Fan Leaderboard:** Top users contributing fans to this sponsor
- **Events Sponsored:** List of events this sponsor is associated with
- **Prize Showcase:** Prizes available from this sponsor with video previews

Sponsor Prizes & Rewards

- **Preset Prize Pool:** Prizes configured by admin for each sponsor tenure
- **Prize Categories:**
 - Top Fan Contributor (individual user who earned most fans)
 - Top Winning Sponsor (sponsor with most event wins)
 - Highest Ranked Sponsor (by composite score)
 - Random Draw (among users who earned >N fans)
- **Prize Display:**
 - Prize name, description, image, video preview
 - Criteria to win
 - Current leader (if applicable)
 - Claim status
- **Prize Distribution:**
 - Auto-awarded at end of tenure
 - Notification to winners
 - Digital certificate + physical prize coordination

Sponsor Archive

- Past sponsors with their tenure period
- Final rankings and fan counts from that tenure
- Winner highlights
- Archived sponsor videos and ads (viewable as legacy content)

Newsletter Section

- Gaming and game development world updates
- Subscribe/Unsubscribe management
- Interaction tracking (opens, clicks) to update user interest profiles
- Newsletter categories:
 - Gaming Industry News
 - Indie Game Spotlights
 - Game Development Tutorials
 - Campus Studio Updates

5.9 F: Hall of Fame

Visibility: Public

- Aesthetic collection of all winners

- Categories:
- League Winners (per sport/esport)
- Highlight Event Winners
- Challenge Legends
- **Sponsor Champions:**
- Top sponsor per semester/year with fan count
- Top individual contributors per sponsor (“MVP” fans earners)
- Sponsor dynasty tracking (consecutive wins)
- Visual timeline of sponsor dominance
- Winner cards: Event name, Winner name/team, Date, Trophy icon, Quote, Sponsor affiliation (if applicable)
- Filter by year, event type, sport, sponsor
- Shareable winner cards

5.10 P: Store Page

Visibility: Authenticated (browse), Authenticated (redeem)

Merchandise Store

- Points redemption for physical/digital merch
- Item cards: Image, name, description, points cost, stock status
- Cart and checkout flow (points deduction)
- Order tracking: Pending → Processing → Shipped → Delivered

Indie Game Support

- Publish and promote games developed by campus studio or partner studios
- Game cards: Trailer, description, download link, points cost (if paid)
- Future: Separate app once marketing scales

Friendly Games Jam Section

- Idea board for game concepts
- Users can pitch ideas under Dev section
- Studio admins (Founder perms) manage submissions
- Upvote/comment system on pitches

Friendly Gaming Section

- Third-party overlay integration for voice calls
- Discord/Steam integration for coordination

5.11 F: Media Page

Visibility: Public (with permission-based filtering)

Layout

- Masonry grid (Pinterest-style) with lazy loading
- Dynamic layout adapting to screen size

Sections

- **Event Albums:** Auto-generated per event (Offside, Waves, etc.)
- **Sponsor Media:** Sponsor videos, ads, and promotional content archive
- **Community Uploads:** User-generated clips/images (moderated)
- **Memories:** “Year in Review” auto-compilation for each user
- **Sponsor Galleries:** Media from sponsor-branded events

Filters

- By event tag
- By sponsor
- By date range
- By uploader
- By media type (photo / video / clip / sponsor_ad)
- By sponsor

Actions

- Download (if permitted)
- Share to external platforms
- Add to “My Memories”
- Report content

Permissions

- Public media: Visible to guests
- Sponsor media: Visible to all (promotional content)
- Private media (from friends-only posts): Visible only to friends
- Event-specific media: Visible based on event visibility
- Sponsor media: Visible to all

Upload Flow

- Via Add Post PopUp → auto-tagged to event if posted from event page
- Camera and gallery access required

5.12 F: Feedback & Contact Us

Visibility: Public

Feedback Ticket System

- Categories: Bug Report, Feature Request, Event Complaint, General
- Severity: Low, Medium, High, Critical
- Description field with rich text
- Anonymous submission toggle

- Attachment support (screenshots, up to 5MB)
- Auto-reply with ticket ID
- Status tracking: Submitted → Under Review → Resolved → Closed

Contact Directory

- Current Coordinators: Name, Role, Email, WhatsApp (masked until clicked to reveal)
- Past Coordinators (Hall of Admin): Legacy team with tenure period, role, and quote
- Quick action buttons: Email, Copy contact, Report issue to specific coordinator

FAQ Section

- Expandable accordion with search functionality
- Sections: Account, Events, Points, Union, Technical, Privacy, Sponsors
- Auto-suggest based on search keywords

5.13 P: Union Page (Internal Workspace)

Visibility: Member, Core, Coordinator, Founder only

Task & Project Management

- **Task Creation:**
 - Quick Add: Overlay accessible from anywhere in app. Minimal fields (title only). Goes to “Unassigned” bucket. For memory-logging without due diligence.
 - Strict Add: Full form with deadline, assignees, event association, priority, description.
- **Task Hierarchy:**
 - Main Tasks
 - Grouped Tasks (task groups)
 - Mini Tasks (sub-tasks)
 - Pathway Tasks: Continuous trackable steps with deadline (e.g., “Make a 2D Game”). Steps appear unplanned initially. Task marked active if updated regularly; otherwise shifted to “Abandoned” for others to pick up.
- **Task Views:**
 - List View
 - Gantt Chart View
 - Calendar View
 - Kanban Board View
- **Task Status:** Active, Abandoned, Completed, On-Hold
- **Task Privacy:** Public (visible to all Union) or Private (assignees only)
- **Task Updates:** Threaded updates per task with timestamp and author
- **Day Notes:** Personal daily notes for members
- **General Notes:** Shared or private note-taking feature

Team Coordination & Scheduling

- **Automated Task Chats:** Assigning users to a task auto-generates a dedicated group chat linked to that task. Task becomes a messaging group.
- **Availability Tracking:** Calendar view for core members to mark free/busy days. Used by task leads for delegation.

- **Google Calendar Sync:**
 - Two-way integration
 - Import academic schedules and personal calendars
 - Export task deadlines and meeting times
 - Push notifications for upcoming meetings

Reminders & Notifications

- **To-Do Reminders:** Overlay reminders for long-term deadlines that appear when opening the app
- **Push Notifications:** Parallel system for deadline alerts (requires background permissions)
- **Priority Flags:** Mark tasks as “Today”, “This Week”, “Upcoming”, “Backlog”

Union Page Workspace Roles (Per Task)

- **Task Leads (Minimum 2):**
 - Full edit freedom on tasks
 - Schedule meetings for all assignees
 - Assign members to sub-tasks
 - Create task documentation
 - Edit task metadata
- **Task OCs (Minimum 2):**
 - Allocate sub-tasks to members
 - Set deadlines for assignees
 - Manage meeting details / set meetings
 - Create/remove temporary sub-OCs (requires Lead approval)
- **Task Members:**
 - Primary execution team
 - Can update task status and add comments
 - Can view linked chat and documents

5.14 P: Users Page (Administration)

Visibility: Coordinator, Founder only

User Management Table

- Complete list of all registered users
- Columns: Name, Email, Role, Status (Active/Disabled), Last Active, Events Count, Join Date, Sponsor Affiliation
- Search by: Email, Name, Phone, Username, Sponsor
- Filters: Role, Status, Date range, Sponsor

Role Management

- **Promotion to Core:** Requires Coordinator approval + reason log
- **Promotion to Coordinator:** Requires Founder approval + 2FA confirmation
- **Demotion / Removal:** Audit trail mandatory. Cannot self-demote.
- **Role Assignment:**
 - User → Member (BGSC crew)
 - Member → Core (BGSC core)

- Core → Coordinator (Current coco)
- Coordinator → Founder (Main project team)

Sponsor Management (Admin Only)

- **Create Sponsor:** Name, logo, description, tenure period, prize pool configuration
- **Edit Sponsor:** Update details, activate/deactivate
- **View Sponsor Stats:** Total fans, affiliated users, events won, rankings
- **Manage Prizes:** Add/edit/remove preset prizes for a sponsor tenure
- **Manage Media:** Upload sponsor videos, ads, and promotional content
 - Hero video upload (background loop)
 - Ad video upload with CTA configuration
 - Story video upload (9:16 format)
 - Media ordering and display scheduling
 - Archive/unarchive media
- **End Tenure:** Close sponsor period, calculate final rankings, trigger prize distribution
- **Sponsor Reports:** Export CSV of all sponsor-related data

Audit & Security

- **Audit Log:** Immutable record of all role changes, event deletions, point modifications, sponsor changes
- **Impersonation View:** Admins can “View as User” for debugging (action logged)
- **Bulk Actions:** Role change, Disable account, Export to CSV
- **Account Actions from Admin:**
 - Disable account (temporary)
 - Remove account (permanent, with data retention policy)
 - Force password reset

6. Popups & Modal Specifications

6.1 Account Action PopUp

Trigger: Clicking profile picture on Status Bar **Layout:** Bottom sheet covering full width, 3/4 screen height

Section 1: Edit Account

- **Status Bar:** Back button (left) → returns to User-Profile Page
- **Fields:**
 - Username (editable)
 - Email (editable, re-verification required)
 - Contact (editable)
 - Password change (does NOT require current password — uses verified session)
- **Interest Fields:** Button opens Interest Fields PopUp with current selections highlighted
- **Newsletters:** Manual subscribe/unsubscribe to categories
- **Socials:** Connect/disconnect Instagram, WhatsApp, Twitch, Discord
- **Sponsor:** Change active sponsor (limited to once per semester, with confirmation)

Section 2: Account Actions

- **Status Bar:** Back button (left) → returns to User-Profile Page
- **Actions:**
 - Disable Account (soft delete, reversible)
 - Remove Account (permanent deletion, data export offered)
 - Account Information Request (GDPR-style data dump including all chat history)
 - Privacy Policy and Terms of Service (view only)

6.2 Interest Fields PopUp

Trigger: First login (Get Started), Account Edit, Manual update prompt **Layout:** Multi-select grid with categories

Categories & Options

- **Sports:** Football, Basketball, Cricket, Badminton, Table Tennis (TT)
- **Esports:** Valorant, CS (Counter-Strike), Tekken, Minecraft, Other Esport Games
- **Gaming Industry:** Story Mode Games, Indie Games
- **Game Development:** Unity, Unreal Engine, Indie Games from Scratch

Behavior

- Current selections highlighted
- Save/Update button
- Weekly/Monthly prompt to update interests (configurable in settings)

6.3 Get Started PopUp (Onboarding)

Trigger: First successful login **Layout:** Sequential pages, must be completed to access full app

Page Sequence

1. **Interest Fields PopUp** (as above)
2. **Sponsor Selection:**
 - Display all active sponsors for current semester/year
 - Sponsor cards with logo, description, current fan count, ranking
 - Mandatory selection (cannot skip)
 - Tooltip: “You can change this once per semester”
3. **Add Friends:**
 - Suggestions based on selected interests and sponsor affiliation
 - Contact list matching (with permission)
 - Interest-based recommendations
4. **Connect Socials:**
 - Instagram
 - WhatsApp (contact card)
 - Twitch
 - Discord

6.4 Make Announcement PopUp

Trigger: Admin/Coordinator action from Home Page or dedicated button **Permissions:** Coordinator, Founder, Core (with permission)

Fields

- Title
- Body (rich text)
- Announcement Type (multi-select tags):
 - BGEC (Visible to all)
 - FitSoc (Visible to all)
 - Airball (Visible to all)
 - Offside (Visible to all)
 - PowerPlay (Visible to all)
 - Around The Net (Visible to all)
 - Deuce (Visible to all)
 - Highlight Events (Visible to all)
 - Teams (Visible to Core, Coordinator, Founder only)
- **WhatsApp Integration:** Each tag maps to a WhatsApp community group API. Auto-sends to respective groups upon publish.
- Send Now / Schedule for Later

6.5 Add Post PopUp

Trigger: FAB on Home Page or Friends Page **Permissions:** Authenticated only (guests redirected to Login) **Layout:** Full-screen sequential tab form, vertically scrollable

Tab 1: Media Selection

- Camera access (live capture)
- Gallery access (select existing)
- Multi-select enabled
- Preview grid with reorder/delete

Tab 2: Post Details

- **Header / Caption**
- **Tags** (event tags, interest tags, sponsor tag)
- **Description**

Tab 3: Privacy & Interaction Controls

- **Likes Section:**
 - Like count visibility toggle (on/off)
 - Likes feature toggle (enabled/disabled entirely)
 - Note: Cannot see WHO liked (computation optimization)
- **Comments Section:**
 - Comments toggle (on/off)
 - Sharing toggle (allowed/not allowed)
 - Comments visibility:

- * Public: Visible to all
- * Protected: Visible only to post author
- * Private: Visible to all EXCEPT post author
- **Post Visibility:**
 - Public: Visible to all, including guests
 - Protected: Visible to all authenticated users (guests excluded)
 - Private: Visible only to user's friends list
 - **3-Tier Story System:**
 - * Non-Judgemental (Close circle)
 - * Close (Friends)
 - * General (Public/All friends)

Tab 4: Background Music (Future Feature)

- Placeholder for audio overlay

6.6 Profile Picture Section

Trigger: Clicking profile picture on User-Profile Page status bar **Layout:** Bottom sheet, 3/4 screen height

- Camera option (live capture)
- Gallery option (select existing)
- Crop/Zoom controls
- Preview before save

6.7 New Event PopUp

Trigger: FAB on Events Page (Admin/Coordinator only) **Layout:** Full-screen panel with custom status bar

- **Status Bar:** Back button (left) only. Returns to Events Page.

Fields

- **Title ***
- **Description ***
- **Ruleset** (PDF upload or external link)
- **Award List** (if any)
- **In-App Points Award:** Boolean toggle
 - If true: Points amount field
- **Leaderboard Required:** Boolean toggle
- **Teamed Event:** Boolean toggle
- **Event Type Dropdown:**

Option A: League (Always has leaderboard)

- **Sports League:**
 - Select sport: Airball, Offside, PowerPlay, Around The Net, Deuce
- **Esports League:**

- Game name (string field)
- **General League Fields:**
 - League format: Round Robin, Upper-Lower Bracket, Direct Elimination, Elimination after N fails
 - Min participants to activate leaderboard (int)
 - Team size (if teamed)
 - Score parameters: Map of <parameter_name: data_type> (int, bool, float)
 - Score normalization toggle:
 - * Lower limit (0)
 - * Upper limit (1000)
 - Auction requirement:
 - * **If Yes:** Auction event details:
 - Venue, Time, Date
 - * **If No:** Participation deadline + League start event:
 - Venue, Time, Date

Option B: Event

- **If needs leaderboard:**
 - Leaderboard format (same options as league)
 - Min participants to activate
 - Team size (if teamed)
 - Score parameters map
 - Score normalization (lower/upper limits)
- **Venue**
- **Date**
- **Time**
- **Award** (if any)

Administration Fields

- **Core People with admin access:** Multi-select user field
- **Deadline:** End time for matches / event closure (auto-closes after last match or deadline)
- **Auction Event Toggle:** If enabled, same auction features as ALL (Auction Leaderboard League)

7. Administration & RBAC

7.1 Global Role Architecture (5 Tiers)

Role	Access Level & Permissions
Founder	Absolute Access. Can modify anything including altering the Founder role itself and assigning highest-level system permissions. Immutable audit trail for all actions.
Coordinator	Full Operational Access. Can manage events, workspaces, users, and sponsors. Cannot alter Founder roles. Can manage and assign Core/Member roles. Can make global announcements.

Table 2 – continued

Role	Access Level & Permissions
Core	Event & Workspace Management. Full access to manage events they are assigned to. Role changing restricted and requires Coordinator permission. Can manage team formations and match data. Can view sponsor stats.
Member (Union)	Restricted Internal Access. Can edit specific event/task details ONLY if explicitly granted access or assigned to that event/task. Can view internal Union Page features. Can view sponsor data.
User	Standard Access. Public frontend access to register for events, use social features, view content, earn/spend points, affiliate with sponsors. No internal capabilities.

7.2 Permission Enforcement

- **API Gateway:** JWT verification + role extraction on every request
- **Service Level:** Endpoint-level role checks against required permission bitmask
- **Field-Level:** Sensitive fields (email, phone) masked in public APIs unless friendship exists or admin role
- **UI-Level:** Buttons/pages hidden based on role (not just API-rejected)

7.3 Audit & Compliance

- All role changes, event deletions, point modifications, sponsor changes, and auction overrides logged immutably
- Audit log entries: Actor ID, Action, Target Type, Target ID, Previous Value, New Value, Timestamp
- Admin impersonation sessions logged with full traceability

8. Event-Driven System Design

8.1 Domain Events Catalog

User Domain Events

```
UserRegistered { user_id, email, username, sponsor_id, timestamp }
UserLoggedIn { user_id, device, ip, timestamp }
UserProfileUpdated { user_id, changed_fields[], timestamp }
UserRoleChanged { user_id, old_role, new_role, changed_by, timestamp }
UserSponsorChanged { user_id, old_sponsor_id, new_sponsor_id, timestamp }
UserDisabled { user_id, reason, disabled_by, timestamp }
UserDeleted { user_id, timestamp }
```

Event Domain Events

```
EventCreated { event_id, title, type, created_by, timestamp }
EventUpdated { event_id, changed_fields[], updated_by, timestamp }
EventRegistrationOpened { event_id, timestamp }
EventRegistrationClosed { event_id, timestamp }
EventStarted { event_id, timestamp }
EventCompleted { event_id, timestamp }
```

```
EventDeleted { event_id, deleted_by, timestamp }
```

Registration Domain Events

```
RegistrationCreated { registration_id, event_id, user_id, role, timestamp }  
RegistrationCancelled { registration_id, event_id, user_id, reason, timestamp }  
TeamCreated { team_id, event_id, captain_id, name, timestamp }  
TeamUpdated { team_id, changed_fields[], updated_by, timestamp }  
TeamMemberAdded { team_id, user_id, added_by, timestamp }  
TeamMemberRemoved { team_id, user_id, removed_by, timestamp }
```

Social Domain Events

```
FriendRequestSent { request_id, requester_id, recipient_id, timestamp }  
FriendRequestAccepted { request_id, requester_id, recipient_id, timestamp }  
FriendRequestRejected { request_id, requester_id, recipient_id, timestamp }  
FriendRemoved { requester_id, recipient_id, timestamp }  
PostCreated { post_id, user_id, visibility, timestamp }  
PostLiked { post_id, user_id, timestamp }  
CommentCreated { comment_id, post_id, user_id, timestamp }
```

Points Domain Events

```
PointsEarned { transaction_id, user_id, amount, source, reference_id, timestamp }  
PointsSpent { transaction_id, user_id, amount, source, reference_id, timestamp }  
PointsRefunded { transaction_id, user_id, amount, source, reference_id, reason, timestamp }
```

Sponsor Domain Events

```
SponsorCreated { sponsor_id, name, tenure_start, tenure_end, created_by, timestamp }  
SponsorActivated { sponsor_id, activated_by, timestamp }  
SponsorDeactivated { sponsor_id, deactivated_by, timestamp }  
UserAffiliated { affiliation_id, user_id, sponsor_id, timestamp }  
UserSponsorChanged { user_id, old_sponsor_id, new_sponsor_id, timestamp }  
FanEarned { user_id, sponsor_id, event_id, amount, reason, timestamp }  
SponsorPrizeAwarded { prize_id, sponsor_id, user_id, prize_title, timestamp }  
SponsorTenureEnded { sponsor_id, final_rank, total_fans, timestamp }
```

Union Domain Events

```
TaskCreated { task_id, title, type, assignees[], created_by, timestamp }  
TaskAssigned { task_id, user_id, assigned_by, timestamp }  
TaskUpdated { task_id, changed_fields[], updated_by, timestamp }  
TaskCompleted { task_id, completed_by, timestamp }  
TaskAbandoned { task_id, reason, timestamp }  
MeetingScheduled { meeting_id, task_id, attendees[], time, timestamp }
```

Auction Domain Events

```
AuctionStarted { auction_id, event_id, player_id, timestamp }
```

```

BidPlaced { bid_id, auction_id, bidder_id, amount, timestamp }
BidClosed { auction_id, winner_id, final_amount, timestamp }
PlayerSold { auction_id, player_id, team_id, amount, timestamp }
PlayerUnsold { auction_id, player_id, timestamp }

```

8.2 Event Consumers by Service

Service	Events Consumed	Actions Taken
Notification Service	All domain events	Routes to appropriate channel (push, email, WhatsApp) based on user preferences
Points Service	EventCompleted, ChallengeCompleted, RegistrationCreated	Awards points, updates balances
Sponsor Service	EventCompleted, UserAffiliated, FanEarned	Updates sponsor fan counts, rankings, prize eligibility
Search Service	EventCreated, EventUpdated, UserRegistered, PostCreated, SponsorCreated	Updates Elasticsearch indices
Audit Service	UserRoleChanged, EventDeleted, PointsEarned, BidPlaced, SponsorTenureEnded	Writes immutable audit records
Analytics Service	All domain events	Aggregates metrics, updates dashboards
Media Service	PostCreated	Processes images/videos, generates thumbnails
Union Service	TaskCreated, TaskAssigned, MeetingScheduled	Updates calendar, creates chat groups

8.3 Event Store & Replay

- **Kafka Retention:** 7 days for hot topics, 30 days for audit topics, 1 year for compliance topics
- **Replay Capability:** New services can hydrate state by replaying events from beginning
- **Event Sourcing Pattern:** Current state = fold of all events (enables time-travel debugging)
- **Snapshotting:** Daily snapshots of aggregate state to optimize replay performance

9. Third-Party Integrations

9.1 Strava (Physical Sports)

- **Data Pulled:** Activity type, distance, duration, calories, pace, route maps
- **Auth:** OAuth2 with read permissions
- **Sync:** Real-time webhooks + daily batch sync for historical data
- **Display:** Activity feed on user profile, weekly stats summary

9.2 Steam (Esports)

- **Data Pulled:** Games owned, playtime (last 2 weeks + total), public achievements, recently played

- **Auth:** OpenID / OAuth2
- **Sync:** Daily sync (Steam API limitations)
- **Display:** Esports activity section on profile, favorite games, total playtime
- **Note:** Requires Steam to be unblocked on campus network

9.3 Google Calendar (Union Scheduling)

- **Data Pulled:** Free/busy slots, existing events
- **Data Pushed:** Task deadlines, meeting invites, event schedules
- **Auth:** OAuth2 with calendar read/write permissions
- **Sync:** Two-way sync via push notifications + daily reconciliation
- **Use Case:** Crew availability tracking, meeting scheduling, deadline management

9.4 WhatsApp Business API (Announcements)

- **Function:** Auto-broadcast announcements to community groups
- **Trigger:** Announcement creation via Make Announcement PopUp
- **Mapping:** Each announcement tag maps to a specific WhatsApp group ID
- **Fallback:** In-app notification if WhatsApp delivery fails
- **Rate Limiting:** Max 1 announcement per tag per hour to prevent spam

9.5 Discord (Community & Gaming)

- **Data Pulled:** Rich presence (what game user is playing), server invites
- **Auth:** OAuth2
- **Display:** Discord handle on profile, “Join our server” links
- **Future:** Voice call integration for Friendly Gaming section

9.6 Instagram / Twitch / Socials

- **Auth:** OAuth2 where available
- **Display:** Social handles on player card, clickable links
- **Data:** No automated data pull; display-only integration

10. Notification System

10.1 Notification Channels

Channel	Use Cases	Tech
In-App	Universal fallback, all action types	Stored in DB, real-time via polling or WebSocket
Push (FCM)	Event reminders, friend requests, task deadlines, auction updates	Firebase Cloud Messaging
Email	Account actions, weekly digest, ticket replies, password reset	SendGrid / AWS SES
WhatsApp	Announcements, event reminders, critical alerts	WhatsApp Business API

Table 4 – continued

Channel	Use Cases	Tech
SMS	Critical alerts only (backup channel)	Twilio / AWS SNS

10.2 Notification Categories & Types

Event Notifications

- Registration opened for [Event Name]
- Reminder: [Event Name] starts in 24 hours
- Reminder: [Event Name] starts in 1 hour
- Results published for [Event Name]
- Team invite received for [Event Name]
- You earned [X] fans for [Sponsor Name]!

Social Notifications

- Friend request received from [Username]
- [Username] accepted your friend request
- [Username] liked your post
- [Username] commented on your post
- [Username] challenged you to [Challenge Name]

Sponsor Notifications

- New sponsor [Name] is now active for this semester!
- Your sponsor [Name] moved up to rank [X]!
- [Sponsor Name] won the semester! You contributed [X] fans.
- You won a prize from [Sponsor Name]: [Prize Title]
- Sponsor tenure ends in [N] days — earn more fans!

Union / Task Notifications

- You were assigned to task: [Task Name]
- Deadline approaching: [Task Name] (24 hours left)
- Deadline approaching: [Task Name] (1 hour left)
- Meeting scheduled: [Meeting Name] at [Time]
- Task [Task Name] marked as abandoned — available for pickup

System Notifications

- Password changed successfully
- points awarded for [Reason]
- Account disabled by administrator
- New login detected from [Device/Browser]

10.3 User Preferences

- Granular toggles per channel per category

- Default: In-App ON, Push ON (except non-urgent), Email OFF (except security), WhatsApp OFF (except announcements if subscribed)
 - Quiet hours: Configurable Do Not Disturb period
-

11. Security, Privacy & Moderation

11.1 Authentication Security

- **JWT:** Access token (15 min expiry) + Refresh token (7 days, rotating)
- **Password Policy:** Min 8 chars, 1 uppercase, 1 number, 1 special char. BCrypt hashing.
- **OAuth2:** Google sign-in with state parameter CSRF protection
- **2FA:** TOTP-based, required for Coordinator promotion and Founder actions
- **Rate Limiting:**
 - Auth endpoints: 5 attempts per 15 minutes per IP
 - General API: 100 requests per minute per user
 - Auction bidding: 1 request per second per user
 - Friend requests: 10 per day
 - Posts: 5 per hour for new users, 20 per hour for established users
 - Sponsor change: 1 per semester

11.2 Data Privacy

- **Encryption:** AES-256 at rest, TLS 1.3 in transit
- **Field Masking:** Email and phone partially masked in public APIs (e.g., r***@gmail.com)
- **Full Access:** Only visible to friends or admin roles
- **GDPR Compliance:**
 - Account Information Request: Complete data export including all chat history, posts, points transactions, sponsor data
 - Right to deletion: Account removal with 30-day grace period
 - Privacy Policy and Terms of Service accessible pre-registration and in Account Actions

11.3 Content Moderation

Auto-Moderation

- **Profanity Filter:** Configurable word list on posts and comments
- **Image Scanning:** AWS Rekognition / Google Vision API for NSFW content detection
- **Spam Detection:** Rate limiting + duplicate content detection
- **Link Filtering:** Auto-flag suspicious URLs

Report System

- **Report Categories:** Spam, Harassment, Cheating, Inappropriate Content, Other
- **Reporter Anonymity:** Optional anonymous reporting
- **Report Queue:** Dedicated moderation queue for Core+ in Union Page
- **Actions:**
 - Hide content pending review
 - Shadow ban (user sees own content, others don't)
 - Temporary ban: 1 day, 7 days, 30 days

- Permanent ban (with appeal option)
- **Ban Appeals:** One appeal per ban. Coordinator review required. Audit logged.

11.4 Auction Security

- **Bid Finality:** Once placed, cannot be retracted. Prevents bid manipulation.
 - **Timer Synchronization:** Server-authoritative countdown. Client displays are estimates.
 - **Purse Validation:** Server-side validation that captain has sufficient funds before accepting bid.
 - **Concurrent Bid Handling:** Atomic database operations with optimistic locking to prevent race conditions.
-

12. Settings & Preferences

Visibility: Authenticated only **Location:** Accessible via Account Action PopUp

12.1 Account Settings

- Change password (requires current password or verified session)
- Two-factor authentication toggle (TOTP setup/reset)
- Download my data (GDPR export, emailed as ZIP)
- Delete account (with confirmation flow and 30-day grace period)

12.2 Privacy Settings

- Profile visibility: Public / Friends Only / Private
- Who can send friend requests: Everyone / Friends of Friends / No One
- Post default visibility: Public / Protected / Private
- Blocked users list (unblock action available)
- Activity status visibility (online/last seen)
- Sponsor visibility: Show/Hide my sponsor affiliation on profile

12.3 Notification Settings

- Granular toggles per channel (In-App, Push, Email, WhatsApp) per category (Event, Social, Sponsor, Union, System)
- Quiet hours configuration (start time, end time, timezone)
- Newsletter email preferences (frequency: daily, weekly, off)
- Sponsor updates toggle (on/off)

12.4 Integration Settings

- Connect / Disconnect Strava
- Connect / Disconnect Steam
- Connect / Disconnect Google Calendar
- Connect / Disconnect Discord
- Re-sync data buttons for each integration

12.5 Appearance Settings

- Theme: Dark / Light / System Default

- Accent color: BGSC Blue / BGEC Purple / FitSoc Green / Custom
 - Font size: Small / Default / Large
 - Reduced motion toggle (accessibility)
-

13. Search & Discovery

13.1 Global Search

- **Placement:** Accessible from Status Bar (search icon or omnibox)
- **Indexed Entities:** Users, Events, Teams, Posts, Announcements, Store Items, Challenges, Sponsors
- **Search Types:**
 - Fuzzy text matching on names and titles
 - Exact match on tags
 - Partial match on descriptions

13.2 Filters & Facets

- **By Type:** User, Event, Team, Post, Announcement, Item, Sponsor
- **By Date:** Today, This Week, This Month, Custom Range
- **By Tag:** Interest tags, Event tags, Sport categories, Sponsor tags
- **By Status:** Upcoming, Ongoing, Past (for events)

13.3 Suggestions & Discovery

- **Recent Searches:** Last 10 searches, clearable
 - **Trending:** Trending events (by registration velocity), trending challenges, trending sponsors
 - **People You May Know:** Based on mutual friends + shared interests + event co-participation + sponsor affiliation
 - **Recommended Events:** ML-based recommendation using interests + past behavior + exploration factor
-

14. Analytics & Success Metrics

14.1 Product Metrics

- **User Acquisition:** Daily/Weekly/Monthly Active Users (DAU/WAU/MAU), registration conversion rate
- **Engagement:** Average session duration, screens per session, posts per user, friend requests per user
- **Event Metrics:** Event registration conversion rate, drop-off rate in registration funnel, team formation rate
- **Retention:** Day-1, Day-7, Day-30 retention cohorts

14.2 Gamification Metrics

- **Points Velocity:** Average points earned per user per week
- **Store Metrics:** Redemption rate, most popular items, stock turnover

- **Challenge Metrics:** Challenge acceptance rate, completion rate by difficulty, Legend completions
- **Sponsor Metrics:**
 - Sponsor affiliation rate (% of users who selected a sponsor)
 - Fan velocity per sponsor
 - Sponsor-driven engagement (events participated by sponsor-affiliated users vs non-affiliated)
 - Prize claim rate

14.3 Operational Metrics (Union)

- **Task Efficiency:** On-time task completion percentage, average task age, abandoned task rate
- **Communication:** Messages per task group, meeting attendance rate
- **Event Execution:** Time from event creation to first registration, admin response time to tickets

14.4 Technical Metrics

- **Performance:** API p95 latency, p99 latency, database query time
- **Reliability:** Uptime percentage, error rate, crash rate (mobile)
- **Infrastructure:** Server CPU/memory, queue depth, cache hit rate

14.5 Reporting Dashboards

- **Coordinator Dashboard:** Event analytics, registration trends, user growth, sponsor rankings
- **Founder Dashboard:** Platform-wide metrics, sponsor ROI, system health
- **Union Dashboard:** Task completion rates, crew availability heatmap, project timelines
- **Sponsor Dashboard:** (For sponsor admins if external) Fan growth, event wins, user engagement

15. Media & Content Specifications

15.1 Supported Media Types

- **Images:** JPG, PNG, WebP (max 10MB, auto-compressed to WebP)
- **Videos:** MP4, MOV (max 50MB, transcoded to multiple resolutions)
 - **Sponsor Videos:** Max 100MB, transcoded to 720p and 1080p, H.264 encoding
 - **Sponsor Ads:** Max 30MB, 15–30 seconds, transcoded to 720p with aggressive compression
 - **Sponsor Stories:** 9:16 aspect ratio, 15–60 seconds, max 20MB
- **Documents:** PDF only (for rulesets, max 5MB)
- **Audio:** MP3 (future feature for post background music)

15.2 Upload Processing Pipeline

1. Client-side validation (type, size)
2. Pre-signed URL generation for direct-to-S3 upload
3. Virus scanning (ClamAV / cloud-native)
4. Image: Resize to thumbnails (150x150), preview (800x800), full resolution
5. Video: Transcode to 480p, 720p, 1080p with H.264 encoding. Generate thumbnail.
 - Sponsor videos: Additional 720p and 1080p variants with watermark option
 - Sponsor ads: Additional compressed 480p variant for mobile bandwidth
 - Sponsor stories: 9:16 crop, 720p variant, poster frame extraction
6. Metadata extraction (EXIF stripping for privacy, dimensions, duration)

7. CDN cache invalidation and distribution

15.3 Storage & Retention

- **User-generated content:** Retained until account deletion (30-day grace)
 - **Event media:** Retained for 2 years post-event, then archived to cold storage
 - **Announcements:** 4 months active, then archived (1 year retention)
 - **Chat history:** Retained per user data request policy. Deleted upon account removal.
 - **Audit logs:** Immutable, 7-year retention
 - **Sponsor data:** Retained for 3 years post-tenure for historical rankings
 - **Sponsor media:** Active sponsor media retained during tenure + 1 year archive. Ads expired after campaign end.
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16. Future Roadmap

16.1 Social Features

- **Stories / Instants:** 24h ephemeral clips with 3-tier sharing groups
- **Dating / Matchmaking:** “Looking for?” feature based on community interests and player cards
- **Relationship Status:** Optional profile field with privacy controls

16.2 Platform Expansion

- **Multi-Campus:** White-label support for other colleges (custom branding, separate databases)
- **Indie Game Store:** Separate standalone app once catalog scales
- **Tournament API:** External tournament organizers can list events on BGSC platform
- **Sponsor Marketplace:** External brands can bid to become sponsors with self-service dashboards

16.3 Advanced Features

- **AI Matchmaking:** ML-based team recommendations using player card data and historical performance
 - **Live Streaming:** Integration with Twitch/YouTube for event broadcasts
 - **VR/AR:** Virtual trophy room in Hall of Fame, AR player cards
 - **Blockchain (Optional):** NFT-based achievement badges for Hall of Fame entries
 - **Predictive Sponsor Analytics:** ML model predicting which sponsor will win based on affiliated user activity
-

17. Phased Development Roadmap

Philosophy

Development follows the “**Platform → Engagement → Operations → Scale**” model. Each phase builds upon the previous, with clear success criteria before proceeding. The MVP answers: “*Will students use a digital hub for campus sports?*”

Total Timeline: 8 Months (32 Weeks)

Buffer: 1.5 Months (6 Weeks)

Active Development: 6.5 Months (26 Weeks)

Phase 0: Foundation (Weeks 1–2)

Goal: Ship nothing user-facing. Build the engine, establish patterns, and set up operational infrastructure. **Duration:** 2 weeks

Deliverables:

Feature	Detail
Infrastructure	CI/CD pipelines (GitHub Actions), staging environment, Docker setup, basic monitoring
Design System	Core component library, color tokens (BGSC/BGEC/FitSoc themes), typography, spacing system
Backend Core	Auth service (JWT + Google OAuth), User service, RBAC middleware, API Gateway with rate limiting
Database	PostgreSQL schema migrations, Redis cache layer
MVVM Scaffold	Base ViewModel classes, observable patterns, repository layer, React Query integration
Mobile Shell	Navigation drawer, status bar, tab system, empty screens, theme switching
Event Bus (MVP)	In-memory event emitter (upgrade to Kafka in Phase 2)

Success Criteria:

- All services deployable via single command
- Auth flow works end-to-end (register → login → token refresh)
- Design system has 15+ reusable components
- Unit test coverage > 50% for all services

Phase 1: MVP — The Public Platform (Weeks 3–10)

Goal: Validate that students will use this for event discovery and registration. Launch with sponsor affiliation from day one. **Target Audience:** All BITS Goa students (Users + Guests). **Duration:** 8 weeks

In Scope:

Feature	Detail	Priority
Auth	Login/Register (email + Google), Forgot password, Guest browsing	P0

Table 6 – continued

Feature	Detail	Priority
Sponsor System v1	Sponsor selection during onboarding, active sponsor display, basic sponsor leaderboard, fan count on event wins	P0
Home Page	Landing intro, Announcements (last 4 months), Public social feed (read-only)	P0
Events	Browse upcoming/ongoing/past events, Solo registration, Basic event details, Post-event fan awards	P0
User Profile	View profile, Edit basic info, Interest fields (onboarding), Public player card with sponsor badge, Sponsor stats	P0
Admin (Coordinator+)	Create DE (Direct Event) and LE (Leaderboard Event), Post announcements, Basic Users table, Sponsor creation/management	P0
Points System	Manual point awards by admin on event participation, User can see balance, Sponsor bonus points	P1
Notifications	In-app only (event reminders, registration confirm, sponsor updates)	P1
Search	Search events and users by name	P1
Hall of Fame v1	Event winners + Sponsor Champions section	P1

Out of Scope (Post-MVP):

- ❌ Team registration (solo only)
- ❌ Auction system
- ❌ Union Page / Task management
- ❌ Friends system (beyond basic search)
- ❌ Store checkout (display only)
- ❌ Strava/Steam integration (manual activity entry)
- ❌ Advanced search filters

- ❌ WhatsApp API (manual announcement copy-paste)
- ❌ Real-time chat (use WhatsApp groups for MVP)
- ❌ Media upload processing pipeline (basic image upload only)
- ❌ Sponsor prizes (track only, distribute manually)

MVP Tech Stack:

Layer	Technology
Mobile App	React Native (Expo)
Web (Admin)	React + Tailwind
Backend	Node.js + NestJS
Database	PostgreSQL (Supabase/Railway)
Cache	Redis
Auth	Supabase Auth / Firebase Auth
Storage	Supabase Storage / Cloudinary
Hosting	Railway/Render (backend) + Vercel (web)
Event Bus	In-memory event emitter (upgrade to Kafka in Phase 2)

MVP Timeline:

Week	Focus
3-4	Auth service + User service + Sponsor service scaffold + Onboarding flow (with sponsor selection)
5-6	Home page (Landing + Announcements + Feed) + Sponsor leaderboard page
7-8	Events CRUD + Registration flow + Basic leaderboard display + Post-event fan award logic
9-10	Player card + Profile (with sponsor badge) + Points (basic) + Hall of Fame v1 + Admin panel + Polish/Bugfix

Success Criteria:

- 500+ registered users in 4 weeks post-launch
- 70% of registered users select a sponsor during onboarding
- 60% of registered users view at least one event
- 20% event registration conversion rate
- Zero critical security issues
- App store rating ≥ 4.0 (if published)

Phase 2: Community & Engagement (Weeks 11–16)

Goal: Drive retention through social features, challenges, and gamification. **Target Audience:** Registered users. **Duration:** 6 weeks

In Scope:

Feature	Detail
Friends System	Add/Remove/Accept/Reject, Search by mutuals, Friend suggestions by interest + sponsor
Social Feed v2	Create posts (camera/gallery), Like/Comment (with visibility controls), 3-tier visibility
Player Cards v2	Customizable tags, shareable image export, “Find similar players”
Challenges	Browse challenges, Accept challenge, Submit proof, Points award
Leaderboards v2	Auto-generated for LE events, Basic score entry by admin, Live updates
Store (Browse)	View merch, Points pricing, “Coming Soon” checkout
Notifications v2	Push notifications (FCM), Email digests
Search v2	Filter by interests, recent achievements, sponsor
Media Pipeline	Full upload processing (resize, transcode, CDN), Media page with albums
Event Bus Upgrade	Migrate from in-memory to Apache Kafka
Sponsor Prizes v1	Preset prize configuration, auto-tracking of leaders, manual prize distribution

Success Criteria:

- 30% of users send at least one friend request
- 15% of users create a post within 30 days
- 25% challenge acceptance rate
- DAU/MAU ratio > 20%
- Sponsor leaderboard viewed by 50% of users weekly

Phase 3: Operations & League Management (Weeks 17–24)

Goal: Make the platform indispensable for the organizing committee. Enable full league and auction support. **Target Audience:** Members, Core, Coordinators. **Duration:** 8 weeks

In Scope:

Feature	Detail
Union Page	Task creation (Quick Add + Strict Add), Gantt/Calendar views, Task assignment, Priority flags
Team Management	Team creation, Join open teams, Invite system, Team captain flow
Leagues (ALL & DLL)	Full league creation, Bracket generation, Match scheduling, Score parameter system
Auction System	Player base price submission, OC override quota (3/7ths), Live bidding interface, Purse management
Union Chat	Auto-generated group chats per task, Task-linked discussions
Calendar Sync	Google Calendar two-way integration for crew availability
Advanced RBAC	Full 5-tier system, Task Leads (2), Task OCs (2), Task Members
Store (Checkout)	Points redemption, Order tracking, Indie game showcase
Hall of Fame v2	Full archive with sponsor dynasty tracking, automated winner archives
Feedback System	Ticket system, FAQ, Contact directory
Sponsor Prizes v2	Auto-distribution of preset prizes at tenure end, digital certificates

Success Criteria:

- 100% of Core team uses Union Page for task management
- 80% task on-time completion rate
- Auction system handles 50+ concurrent bidders without lag
- Zero data loss in Union operations
- First sponsor tenure completes successfully with prize distribution

Phase 4: Integrations & Polish (Weeks 25–28)

Goal: Differentiate the platform with external integrations and production polish. **Target Audience:** Power users. **Duration:** 4 weeks

In Scope:

Feature	Detail
External Integrations	Strava activity sync, Steam library sync, Discord rich presence
WhatsApp API	Announcement auto-broadcast to community groups

Table 11 – continued

Feature	Detail
Media Memories	“Year in Review” auto-compilation, Event highlight reels
Advanced Analytics	Coordinator dashboard for event analytics, User cohort retention, Sponsor ROI dashboard
Performance Optimization	Image compression, CDN, query optimization, bundle splitting
Accessibility	WCAG 2.1 AA compliance audit
Security Hardening	Penetration testing, security audit
Sponsor Analytics	Predictive sponsor rankings, engagement heatmaps

Success Criteria:

- 40% of users connect at least one external integration
- 20% of users view their “Year in Review”
- API p95 latency < 200ms
- Zero critical vulnerabilities in penetration test
- Platform ready for next semester sponsor rollout

Phase 5: Buffer & Contingency (Weeks 29–32)

Goal: Absorb delays, fix critical bugs, and prepare for launch/marketing. **Duration:** 4 weeks (1 month buffer) — **Note:** Total buffer is 1.5 months (6 weeks). The remaining 2 weeks are distributed as flex time across previous phases.

Activities:

- Bug fixes from Phase 4 testing
- Performance tuning based on load testing
- Marketing material creation (app store screenshots, demo videos)
- Campus ambassador onboarding
- Soft launch to 100 beta users
- Final security review
- Documentation completion (API docs, user guides, admin guides)

Success Criteria:

- All P0 bugs resolved
- Load test passed: 1000 concurrent users
- Beta user NPS > 50
- App store submission ready (if applicable)

18. Appendix: Glossary

Term	Definition
BGSC	BITS Goa Sports Community
BGEC	BITS Goa Esports Community
FitSoc	Fitness Society
Union Page	Internal workspace for organizing committees
Player Card	Shareable digital card showcasing user gaming/sports profile
Instants	Ephemeral story-like posts (future feature)
ALL	Auction Leaderboard League
DLL	Direct Leaderboard League
LE	Leaderboard Event
DE	Direct Event (no leaderboard)
OC	Organizing Committee member
K Multiplier	Auction purse calculation factor: $\text{Pool} = K \times \Sigma \text{Base Prices}$
Quick Add	Rapid task creation without full details (memory logging)
Pathway Task	Multi-step task with deadline and progress tracking
3/7ths Quota	OC override limit for player base prices in auction
Non-Judgemental	Closest privacy tier for stories/posts
Protected	Visible to all authenticated users, hidden from guests
FCM	Firebase Cloud Messaging
TOTP	Time-based One-Time Password (2FA)
GDPR	General Data Protection Regulation
CSRF	Cross-Site Request Forgery
RBAC	Role-Based Access Control
MVP	Minimum Viable Product
SLA	Service Level Agreement
CDN	Content Delivery Network
PWA	Progressive Web App
EXIF	Exchangeable Image File Format (metadata)
EDA	Event-Driven Architecture
MVVM	Model-View-ViewModel
CQRS	Command Query Responsibility Segregation
Kafka	Apache Kafka (distributed event streaming platform)
Avro	Data serialization system for Kafka schemas
Fan Count	Virtual points earned by users for their sponsor when winning events
Sponsor Tenure	A semester or year period during which a sponsor is active
Sponsor Dynasty	Consecutive tenure wins by the same sponsor

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